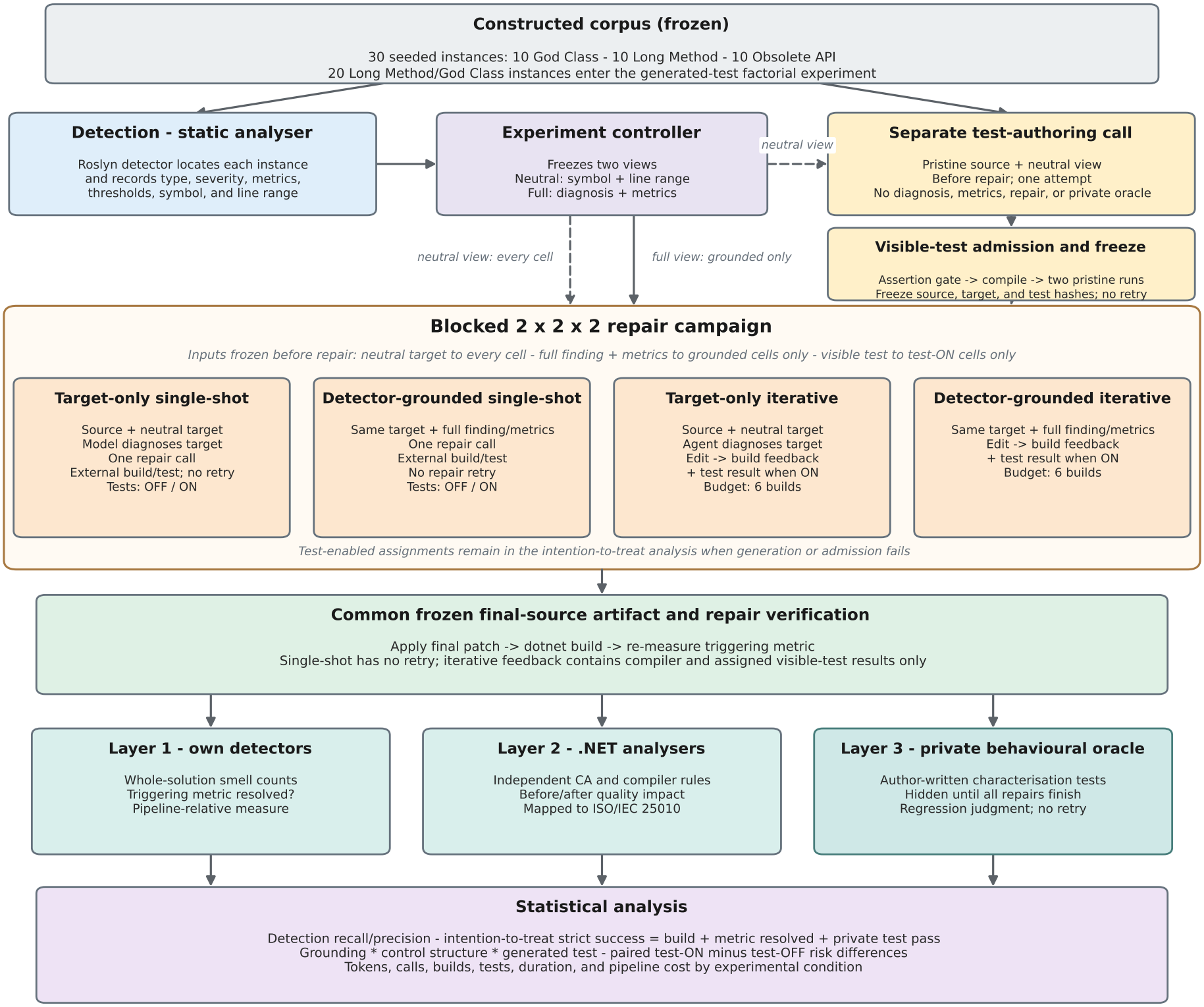
Constructed corpus (frozen)
30 seeded instances: 10 God Class - 10 Long Method - 10 Obsolete API
20 Long Method/God Class instances enter the generated-test factorial experiment
Detection - static analyser
Roslyn detector locates each instance and records type, severity, metrics, thresholds, symbol, and line range
Experiment controller
Freezes two views
Neutral: symbol + line range
Full: diagnosis + metrics
neutral view
Separate test-authoring call
Pristine source + neutral view
Before repair; one attempt
No diagnosis, metrics, repair, or private oracle
neutral view: every cell
full view: grounded only
Visible-test admission and freeze
Assertion gate -> compile -> two pristine runs
Freeze source, target, and test hashes; no retry
Blocked 2 x 2 x 2 repair campaign
Inputs frozen before repair: neutral target to every cell - full finding + metrics to grounded cells only - visible test to test-ON cells only
Target-only single-shot
Source + neutral target
Model diagnoses target
One repair call
External build/test; no retry
Tests: OFF / ON
Detector-grounded single-shot
Same target + full finding/metrics
One repair call
External build/test
No repair retry
Tests: OFF / ON
Target-only iterative
Source + neutral target
Agent diagnoses target
Edit -> build feedback
+ test result when ON
Budget: 6 builds
Detector-grounded iterative
Same target + full finding/metrics
Edit -> build feedback
+ test result when ON
Budget: 6 builds
Test-enabled assignments remain in the intention-to-treat analysis when generation or admission fails
Common frozen final-source artifact and repair verification
Apply final patch -> dotnet build -> re-measure triggering metric
Single-shot has no retry; iterative feedback contains compiler and assigned visible-test results only
Layer 1 - own detectors
Whole-solution smell counts
Triggering metric resolved?
Pipeline-relative measure
Layer 2 - .NET analysers
Independent CA and compiler rules
Before/after quality impact
Mapped to ISO/IEC 25010
Layer 3 - private behavioural oracle
Author-written characterisation tests
Hidden until all repairs finish
Regression judgment; no retry
Statistical analysis
Detection recall/precision - intention-to-treat strict success = build + metric resolved + private test pass
Grounding * control structure * generated test - paired test-ON minus test-OFF risk differences
Tokens, calls, builds, tests, duration, and pipeline cost by experimental condition